1.CREATE AN EXPOSURE OF WEB APPLICATION AND TOOLS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="EX1.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
      <div>

     <asp:Label ID="Label1" runat="server" Font-Size="Large"></asp:Label>
  </div>
   </form>
</body>
</html>
```

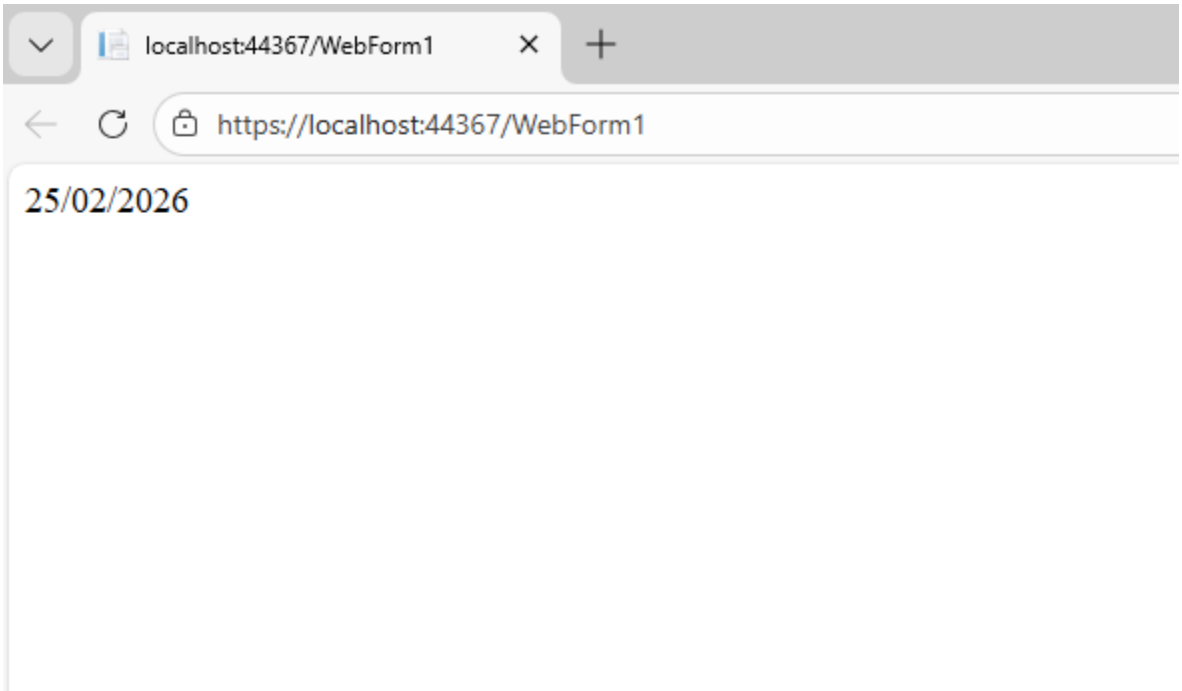
BEHIND THE CODE

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace EX1
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {
         Label1.Text = DateTime.Now.ToString("dd/MM/yyyy");
      }

   }
}
```

OUTPUT


localhost:44367/WebForm1
https://localhost:44367/WebForm1
25/02/2026

2.IMPLEMENT THE HTML CONTROLS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="program_2_html_controls.WebForm1" %>
            <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
                  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
   <style type="text/css">
      .style1
      {
         width: 100%;
      }
      .style2
      {
         height: 26px;
      }
   </style>
</head>
<body>
   <form id="form1" runat="server">
   <div>

      <table class="style1">
         <tr>
            <td>
               Name:</td>
            <td>

<asp:TextBox ID="txtname" runat="server"></asp:TextBox>
            </td>
         </tr>
         <tr>
            <td>
               Street:</td>
            <td>

<asp:TextBox ID="txtstreet" runat="server"></asp:TextBox>
            </td>
         </tr>
         <tr>
            <td>
               City:</td>
```

```
            <td>

<asp:TextBox ID="txtcity" runat="server"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>
                State:</td>
            <td>

<asp:TextBox ID="txtstate runat="server"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td class="style2">
            </td>
            <td class="style2">
            </td>
        </tr>
        <tr>
            <td>
                 </td>
            <td id="displayrow" runat="server">
                 </td>
        </tr>
    </table>

  </div>
  <asp:Button ID="btnclick" runat="server" Text="Click" OnClick="btnclick_click" />
  <p>

<asp:Label ID="lbldisplay"runat="server"></asp:Label>
  </p>
  </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace program_2_html_controls
   {
   public partial class WebForm1:System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {

      }

      protected void btnclick_click(object sender, EventArgs e)
      {
         string str = "";
         str += txtname.Text + "<br />";
         str += txtstreet.Text + "<br />";
         str += txtcity.Text + "<br />";
         str += txtstate.Text + "<br />";
         displayrow.InnerHtml = str;

      }
   }
}
```

OUTPUT

Name: shakkira A

Street: periyar nagar virddhachalam

City: vriddhachalam

State: Tamil Nadu

shakkira A
periyar nagar virddhachalam
vriddhachalam
Tamil Nadu

Click

3.IMPLEMENT THE SERVER CONTROLS

```
<%@ Page Title="Home Page" Language="C#" MasterPageFile="~/Site.master"
AutoEventWireup="true"
   CodeBehind="Default.aspx.cs" Inherits="server_controls._Default" %>

<asp:Content ID="HeaderContent" runat="server" ContentPlaceHolderID="HeadContent">
</asp:Content>
<asp:Content ID="BodyContent" runat="server" ContentPlaceHolderID="MainContent">
   <p>
      <br />
      <asp:Label ID="lblname" runat="server" Text="Enter your name"></asp:Label>
      <asp:TextBox ID="txtname" runat="server"></asp:TextBox>
   </p>
   <p>
      <asp:Button ID="btnsubmit" runat="server" Text="Greet me" onclick="btnsubmit_Click"
/>
      <asp:Label ID="lblgreeting" runat="server"></asp:Label>
   </p>
   <p>
   </p>
   <p>
   </p>
   <p>
   </p>
</asp:Content>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace server_controls
{
   public partial class _Default : System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {

      }

      protected void btnsubmit_Click(object sender, EventArgs e)
      {
         lblgreeting.Text = "Hello, " + txtname.Text + "!";
      }
   }
}
```

OUTPUT:

Enter your name A.shakkira

Greet me Hello, A.shakkira!

4.WEB APPLICATION USING WEB CONTROLS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="prgram_4_web_controls.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
   <div>

      <asp:Label ID="lblname" runat="server" Text="Student Name:"></asp:Label>
      <asp:TextBox ID="txtname" runat="server"></asp:TextBox>
      <br />
      <br />
      <asp:Label ID="lbldepartment" runat="server" Text="Department:"></asp:Label>
      <asp:DropDownList ID="ddldepartment" runat="server">
         <asp:ListItem>Computer science</asp:ListItem>
         <asp:ListItem>Electric</asp:ListItem>
         <asp:ListItem>Mechanical</asp:ListItem>
      </asp:DropDownList>
      <br />
      <br />
      <asp:Label ID="lblmode" runat="server" Text="Mode of learning:"></asp:Label>
      <asp:RadioButtonList ID="rblmode" runat="server">
         <asp:ListItem>online</asp:ListItem>
         <asp:ListItem>offline</asp:ListItem>
      </asp:RadioButtonList>
      <br />
      <asp:Label ID="lbllanguage" runat="server" Text="Programming language
known:"></asp:Label>
      <br />
      <asp:CheckBoxList ID="cbllanguage" runat="server">
         <asp:ListItem>C</asp:ListItem>
         <asp:ListItem>Java</asp:ListItem>
         <asp:ListItem>Python</asp:ListItem>
         <asp:ListItem>C++</asp:ListItem>
      </asp:CheckBoxList>
      <br />
```

```
        <asp:Button ID="btnsubmit" runat="server" Text="Register"
OnClick="btnsubmit_Click"/>

        <br />
        <br />
        <asp:Label ID="lbldisplay" runat="server"></asp:Label>

    </div>
    </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace prgram_4_web_controls
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {

      }

      protected void btnsubmit_Click(object sender, EventArgs e)
      {
         String name = txtname.Text.Trim();
         String department = ddldepartment.SelectedValue;
         String mode = rblmode.SelectedItem != null ? rblmode.SelectedItem.Text : "Not
Selected";
         String language = "";
         foreach (ListItem item in cbllanguage.Items)
         {
            if (item.Selected)
            {
               language += item.Text + ",";
            }
         }
         if (language.EndsWith(","))
         {
            language = language.Substring(0, language.Length - 2);
         }
         if (String.IsNullOrEmpty(name))
         {
            lbldisplay.Text = "Please select a department.";
            lbldisplay.ForeColor = System.Drawing.Color.Red;
            return;
         }
         lbldisplay.ForeColor = System.Drawing.Color.Blue;
         lbldisplay.Text = "<b>StudentName:</b>+name+<br/>" +
"<b>Department:</b>+department+<br/>" + "<b>Mode of learning:</b>+mode+<br/>" +
"<b>Programming language known:</b>+language";
```

```
        }
    }
}
```

OUTPUT

Student Name: A.shakkira

Department: Computer science

Mode of learning:

- ○ online
- ◉ offline

Programming language known:

- ☐ C
- ☐ Java
- ☑ Python
- ☐ C++

Register

StudentName:+name+
Department:+department+
Mode of learning:+mode+
Programming language known:+language

5.WEB APPLICATION USING LIST CONTROLS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="program_5_list_controls.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
   <asp:Label ID="lbllanguage" runat="server"
      Text="Select your programming language:"></asp:Label>
   <br />
   <asp:DropDownList ID="ddllanguage" runat="server">
      <asp:ListItem>Python</asp:ListItem>
      <asp:ListItem>Java</asp:ListItem>
      <asp:ListItem>C++</asp:ListItem>
   </asp:DropDownList>
   <br />
   <br />
   <asp:Label ID="lblhobbies" runat="server" Text="Select your hobbies:"></asp:Label>
   <br />
   <asp:CheckBoxList ID="cblhobbies" runat="server">
      <asp:ListItem>reading</asp:ListItem>
      <asp:ListItem>gaming</asp:ListItem>
      <asp:ListItem>traveling</asp:ListItem>
   </asp:CheckBoxList>
   <br />
   <asp:Label ID="lblcompany" runat="server" Text="Select your company:"></asp:Label>
   <asp:RadioButtonList ID="rblcountry" runat="server">
      <asp:ListItem>India</asp:ListItem>
      <asp:ListItem>USA</asp:ListItem>
      <asp:ListItem>UK</asp:ListItem>
   </asp:RadioButtonList>
   <asp:Button ID="btnsubmit" runat="server" onclick="btnsubmit_Click"
      Text="Submit" />
   <p>
      <asp:Label ID="lblresult" runat="server"></asp:Label>
   </p>
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace program_5_list_controls
{
   public partial class WebForm1:System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {

      }

      protected void btnsubmit_Click(object sender, EventArgs e)
      {
         String selectedLanguage = ddllanguage.SelectedItem.Text.Trim();
         String selectedHobbies = "";
         foreach (ListItem item in cblhobbies.Items)
         {
            if (item.Selected)
            {
               selectedHobbies += item.Text.Trim() + ",";
            }
         }
         if (selectedHobbies.EndsWith(","))
         {
            selectedHobbies = selectedHobbies.Substring(0, selectedHobbies.Length - 2);
         }
         String selectedCountry = rblcountry.SelectedItem != null ?
rblcountry.SelectedItem.Text.Trim() : "Not selected";
         lblresult.Text = "<b>Selected programming language:</b>+selectedLanguage<br/>" +
"<b>Selected Hobbies:</b>+selectedHobbies<br/>" + "<b>Selected
country:</b>+selectedCountry";
      }
   }
}
```

OUTPUT

Select your programming language:

Python

Select your hobbies:

- [ ] reading
- [x] gaming
- [x] traveling

Select your company:

- (•) India
- ( ) USA
- ( ) UK

Submit

Selected programming language:+selectedLanguage
Selected Hobbies:+selectedHobbies
Selected country:+selectedCountry

6.WEB PAGE DESIGN RICH CONTROLS USING VALIDATE CONTROLS WITH FILE CONCEPT

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="program_6_rich_controls.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
   <div>

      <asp:Label ID="lblName" runat="server" Text="Name:"></asp:Label>
      <asp:TextBox ID="txtName" runat="server" CssClass="rich-input"></asp:TextBox>
      <asp:RequiredFieldValidator ID="rfvName" runat="server"
         ControlToValidate="txtName" ErrorMessage="Name Required"
ForeColor="Red"></asp:RequiredFieldValidator>
      <br />
      <br />
      <asp:Label ID="lblEmail" runat="server" Text="Email:"></asp:Label>
      <asp:TextBox ID="txtEmail" runat="server" CssClass="rich-input"></asp:TextBox>
      <asp:RequiredFieldValidator ID="rfvEmail" runat="server"
         ControlToValidate="txtEmail" ErrorMessage="Email Required"
ForeColor="Red"></asp:RequiredFieldValidator>
      <br />
      <asp:RegularExpressionValidator ID="revEmail" runat="server"
         ControlToValidate="txtEmail" ErrorMessage="Invalid Email" ForeColor="Red"
         ValidationExpression="\w+([-+.']\w+)*@\w+([-.]\w+)*\.\w+([-
.]\w+)*"></asp:RegularExpressionValidator>
      <br />
      <br />
      Salary:<asp:TextBox ID="txtSal" runat="server"></asp:TextBox>
      <asp:RangeValidator ID="rvSal" runat="server" BackColor="White"
         ControlToValidate="txtSal" ErrorMessage="Salary must from 10000 to 20000"
         ForeColor="#FF0066" MaximumValue="20000"
MinimumValue="10000"></asp:RangeValidator>
      <br />
      <br />
      Data of joining:<asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
      <asp:Button ID="btnSave" runat="server" CssClass="rich-button"
```

```
            Text="Save to File" OnClick="btnSave_Click" />
        <br />
        <br />
        <asp:Label ID="lblMessage" runat="server" ForeColor="Black"></asp:Label>
        <br />

    </div>
    </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace program_6_rich_controls
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {

      }

      protected void btnSave_Click(object sender, EventArgs e)
      {
         if (Page.IsValid)
         {
            String userData = "Name: {txtName.Text}, Email: {txtEmail.Text}, Salary:
{txtSalary.Int}, Date: {DateTime.Now}\n";

            //File concept:Append to data.txt
            String filePath = Server.MapPath("~/data.txt");
            System.IO.File.AppendAllText(filePath, userData);
            lblMessage.Text = "Data saved to file successfully!";
            txtName.Text = "";// clear inputs
            txtEmail.Text = "";
            txtSal.Text = "";
         }
         else
         {
            lblMessage.Text = "Please fix validation errors.";
         }
      }
   }
}
```

OUTPUT

Name: A.shakkira

Email: shakkiraabdulravuf@gmail.c

Salary: 15000

Data of joining:

≤		February 2026				≥
Sun	**Mon**	**Tue**	**Wed**	**Thu**	**Fri**	**Sat**
25	26	27	28	29	30	31
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
1	2	3	4	5	6	7

Save to File

Data saved to file successfully!

7.WEB APPLICATION USING DATA CONTROLS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="program_7_grid_view.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">

   <asp:GridView ID="gvdata" runat="server" AutoGenerateColumns="False"
      CellPadding="15">
      <Columns>
         <asp:BoundField DataField="ID" HeaderText="ID" />
         <asp:BoundField DataField="Name" HeaderText="Name" />
         <asp:BoundField DataField="City" HeaderText="City" />
      </Columns>
   </asp:GridView>
   <br />

   <div>

   </div>
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data;

namespace program_7_grid_view
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {
         if(!IsPostBack)
         {
            gvdata.DataSource=GetSampleData();
            gvdata.DataBind();
         }
      }
      private List<Employee>GetSampleData()
      {
         return new List<Employee>
         {
            new Employee { ID=1,Name="John",City="NYC"},
            new Employee { ID=2,Name="Jane",City="LA"},
            new Employee { ID=3,Name="Mike",City="Chicago"}
         };
      }
   }
}
public class Employee
{
   public int ID {get;set;}
   public string Name {get;set;}
   public string City {get;set;}
}
```

OUTPUT:

ID	Name	City
1	John	NYC
2	Jane	LA
3	Mike	Chicago

8.DATA BINDING WITH WEB CONTROLS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="program_8_databinding.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
   <div>
   <h2>DropDownList</h2>
      <asp:DropDownList ID="ddlnames" runat="server">
      </asp:DropDownList>
      <br />
      <br />
      <br />
    <h2>ListBox</h2>

      <asp:ListBox ID="lstnames" runat="server"></asp:ListBox>
      <br />
      <br />
   <h2>GridView</h2>
         <asp:GridView ID="gvnames" runat="server" AutoGenerateColumns="False"
          AutoGenerateSelectButton="True">
      </asp:GridView>


     <h2>Selected Item</h2>
 <asp:Label ID="lblresult" runat="server"></asp:Label>
 </div>
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data;

namespace program_8_databinding
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                BindData();
            }
        }
        void BindData()
        {
            DataTable dt = new DataTable();
            dt.Columns.Add("ID");
            dt.Columns.Add("Name");

            dt.Rows.Add("1", "Ravi");
            dt.Rows.Add("2", "Anitha");
            dt.Rows.Add("3", "Kumar");
            dt.Rows.Add("4", "Divya");

            // DropDownList
            ddlnames.DataSource = dt;
            ddlnames.DataTextField = "Name";
            ddlnames.DataValueField = "ID";
            ddlnames.DataBind();

            //ListBox
            lstnames.DataSource = dt;
            lstnames.DataTextField = "Name";
            lstnames.DataValueField = "ID";
            lstnames.DataBind();

            //GridView
            gvnames.DataSource = dt;
            gvnames.DataBind();
```

```
    }
    protected void ddlnames_SelectedIndexChanged(object sender,EventArgs e)
    {
        lblresult.Text = "Selected from DropDownList:"+ ddlnames.SelectedItem.Text;
    }
    protected void lstnames_SelectedIntexChanged(object sender,EventArgs e)
    {
        lblresult.Text = "Selected from ListBox:" + lstnames.SelectedItem.Text;
    }
    protected void gvnames_SelectedIndexChanged(object sender,EventArgs e)
    {
        lblresult.Text = "Selected from GridView:" + gvnames.SelectedRow.Cells[2].Text;
    }
    }
}
```

OUTPUT:

DropDownList

ListBox

GridView

	ID	Name
Select	1	Ravi
Select	2	Anitha
Select	3	kumar
Select	4	Divya

Selected Item

Selected from GridView:Ravi

9.DATA BINDING WITH DATA CONTROLS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="program_9_datacontrols.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
   <asp:GridView ID="GridView1" runat="server" AutoGenerateColumns="True"
      onselectedindexchanged="GridView1_SelectedIndexChanged">
   </asp:GridView>
   <br />
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data;

namespace program_9_datacontrols
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Page_Load(object sender, EventArgs e)
      {
                  if (!IsPostBack)
         {
            BindGrid();
         }
      }
      private void BindGrid()
      {
         DataTable dt = new DataTable();
         dt.Columns.Add("ID");
         dt.Columns.Add("Name");
         dt.Columns.Add("Course");

         dt.Rows.Add("1", "Alice", "Math");
         dt.Rows.Add("2", "Bob", "Physics");
         dt.Rows.Add("3", "Charlie", "Chemistry");
         GridView1.DataSource = dt;
         GridView1.DataBind();
      }

      protected void GridView1_SelectedIndexChanged(object sender, EventArgs e)
      {
         // Just a placeholder: can be used if selection is enabled
      }
   }
}
```

OUTPUT

ID	Name	Course
1	Alice	Math
2	Bob	Physics
3	Charlie	Chemistry

10. DATABASE APPLICATION PERFORM INSERT, UPDATE AND DELETE OPERATIONS

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
Inherits="WebApplication1._Default" %>

<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head id="Head1" runat="server">
   <title>CRUD Operations - ASP.NET</title>
</head>
<body>
   <form id="form1" runat="server">
      <h2>CRUD Operations</h2>

      <!-- User Input Section -->
      <label for="txtId">ID:</label>
      <asp:TextBox ID="txtId" runat="server" Placeholder="User ID" /><br /><br />
      <label for="txtName">Name:</label>
      <asp:TextBox ID="txtName" runat="server" Placeholder="Name" /><br /><br />
      <label for="txtEmail">Email:</label>
      <asp:TextBox ID="txtEmail" runat="server" Placeholder="Email" /><br /><br />

      <!-- Buttons for CRUD Operations -->
      <asp:Button ID="btnInsert" runat="server" Text="Insert User" OnClick="btnInsert_Click"
/><br /><br />
      <asp:Button ID="btnUpdate" runat="server" Text="Update User"
OnClick="btnUpdate_Click" /><br /><br />
      <asp:Button ID="btnDelete" runat="server" Text="Delete User"
OnClick="btnDelete_Click" /><br /><br />

      <!-- Display Results -->
      <asp:Label ID="lblMessage" runat="server" Text="" ForeColor="Green" /><br /><br />
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;

namespace WebApplication1
{
    public partial class _Default : System.Web.UI.Page
    {
        // List to hold users (In-memory, for demonstration purposes)
        private static List<User> users = new List<User>();

        protected void Page_Load(object sender, EventArgs e)
        {
            // Optionally, you can load existing data or handle any pre-load logic here
        }

        // Insert user into the list
        protected void btnInsert_Click(object sender, EventArgs e)
        {
            int id;
            if (int.TryParse(txtId.Text, out id) && !string.IsNullOrEmpty(txtName.Text) &&
!string.IsNullOrEmpty(txtEmail.Text))
            {
                User newUser = new User
                {
                    Id = id,
                    Name = txtName.Text,
                    Email = txtEmail.Text
                };
                users.Add(newUser);
                lblMessage.Text = "User inserted successfully!";
                lblMessage.ForeColor = System.Drawing.Color.Green;
            }
            else
            {
                lblMessage.Text = "Please provide valid inputs!";
                lblMessage.ForeColor = System.Drawing.Color.Red;
            }
        }

        // Update user information
        protected void btnUpdate_Click(object sender, EventArgs e)
        {
            int id;
            if (int.TryParse(txtId.Text, out id))
            {
```

```
        var userToUpdate = users.Find(u => u.Id == id);
        if (userToUpdate != null)
        {
            userToUpdate.Name = txtName.Text;
            userToUpdate.Email = txtEmail.Text;
            lblMessage.Text = "User updated successfully!";
            lblMessage.ForeColor = System.Drawing.Color.Green;
        }
        else
        {
            lblMessage.Text = "User not found!";
            lblMessage.ForeColor = System.Drawing.Color.Red;
        }
    }
    else
    {
        lblMessage.Text = "Please provide a valid ID!";
        lblMessage.ForeColor = System.Drawing.Color.Red;
    }
}

// Delete user from the list
protected void btnDelete_Click(object sender, EventArgs e)
{
    int id;
    if (int.TryParse(txtId.Text, out id))
    {
        var userToDelete = users.Find(u => u.Id == id);
        if (userToDelete != null)
        {
            users.Remove(userToDelete);
            lblMessage.Text = "User deleted successfully!";
            lblMessage.ForeColor = System.Drawing.Color.Green;
        }
        else
        {
            lblMessage.Text = "User not found!";
            lblMessage.ForeColor = System.Drawing.Color.Red;
        }
    }
    else
    {
        lblMessage.Text = "Please provide a valid ID!";
        lblMessage.ForeColor = System.Drawing.Color.Red;
    }
}
```

```
        // Simple User class to represent a user
        public class User
        {
            public int Id { get; set; }
            public string Name { get; set; }
            public string Email { get; set; }
        }
    }
}
```

OUTPUT

CRUD Operations

ID: 12345

Name: RANITHA

Email: ranithaelumalai@gmail.com

Insert User

Update User

Delete User

User inserted successfully!

11.DATABASE APPLICATIONPERFORM INSERT , UPDATE , AND DELETE OPERATION ,EDIT PAGING AND SORTING OPERATION

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="Web11.WebForm1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
   <title>ASP.NET CRUD with GridView</title>
</head>
<body>
   <form id="form1" runat="server">
      <h2>Employee Management System</h2>

      <asp:GridView ID="GridView1" runat="server" AutoGenerateColumns="False"
         DataKeyNames="Id" AllowPaging="True" AllowSorting="True"
         PageSize="5"
         OnPageIndexChanging="GridView1_PageIndexChanging"
         OnRowEditing="GridView1_RowEditing"
         OnRowUpdating="GridView1_RowUpdating"
         OnRowCancelingEdit="GridView1_RowCancelingEdit"
         OnRowDeleting="GridView1_RowDeleting"
         OnSorting="GridView1_Sorting">

         <Columns>
            <asp:BoundField DataField="Id" HeaderText="ID" ReadOnly="True"
SortExpression="Id" />
            <asp:BoundField DataField="Name" HeaderText="Name" SortExpression="Name" />
            <asp:BoundField DataField="Age" HeaderText="Age" SortExpression="Age" />
            <asp:BoundField DataField="Department" HeaderText="Department"
SortExpression="Department" />
            <asp:CommandField ShowEditButton="True" ShowDeleteButton="True" />
         </Columns>
      </asp:GridView>

      <br /><hr /><br />
      <h3>Add New Employee</h3>
      <asp:Label ID="lblMessage" runat="server" ForeColor="Green" /><br />
      Name: <asp:TextBox ID="txtName" runat="server" /><br />
      Age: <asp:TextBox ID="txtAge" runat="server" /><br />
      Department: <asp:TextBox ID="txtDept" runat="server" /><br />
      <asp:Button ID="btnAdd" runat="server" Text="Add" OnClick="btnAdd_Click" />
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Data;
using System.Data.SqlClient;

namespace Web11
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        static string SortExpression = "Id ASC";

        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                BindGrid();
            }
        }

        private void BindGrid()
        {
            using (SqlConnection con = new SqlConnection(/* Your connection string here */))
            {
                string query = "SELECT * FROM Employees ORDER BY " + SortExpression;
                SqlDataAdapter da = new SqlDataAdapter(query, con);
                DataTable dt = new DataTable();
                GridView1.DataSource = dt;
                GridView1.DataBind();
            }
        }

        protected void btnAdd_Click(object sender, EventArgs e)
        {
            if (txtName.Text == "" || txtAge.Text == "")
            {
                return;
            }

            using (SqlConnection con = new SqlConnection(/* Your connection string here */))
            {
                string query = "INSERT INTO Employees (Name, Age) VALUES (@Name, @Age)";
```

```
            SqlCommand cmd = new SqlCommand(query, con);
            cmd.Parameters.AddWithValue("@Name", txtName.Text.Trim());
            cmd.Parameters.AddWithValue("@Age", Convert.ToInt32(txtAge.Text.Trim()));

            con.Open();
            cmd.ExecuteNonQuery();
            con.Close();

            txtName.Text = txtAge.Text = "";
            BindGrid();
        }
    }

    protected void GridView1_PageIndexChanging(object sender,
System.Web.UI.WebControls.GridViewPageEventArgs e)
    {
        GridView1.PageIndex = e.NewPageIndex;
        BindGrid();
    }

    protected void GridView1_RowEditing(object sender,
System.Web.UI.WebControls.GridViewEditEventArgs e)
    {
        GridView1.EditIndex = e.NewEditIndex;
        BindGrid();
    }

    protected void GridView1_RowCancelingEdit(object sender,
System.Web.UI.WebControls.GridViewCancelEditEventArgs e)
    {
        GridView1.EditIndex = -1;
        BindGrid();
    }

    protected void GridView1_RowUpdating(object sender,
System.Web.UI.WebControls.GridViewUpdateEventArgs e)
    {
        int id = Convert.ToInt32(GridView1.DataKeys[e.RowIndex].Value);
        string name =
((System.Web.UI.WebControls.TextBox)GridView1.Rows[e.RowIndex].Cells[1].Controls[0]).T
ext;
        string ageText =
((System.Web.UI.WebControls.TextBox)GridView1.Rows[e.RowIndex].Cells[2].Controls[0]).T
ext;

        int age;
```

```
        if (!int.TryParse(ageText, out age))
        {
            return;
        }

        using (SqlConnection con = new SqlConnection(/* Your connection string here */))
        {
            string query = "UPDATE Employees SET Name=@Name, Age=@Age WHERE
Id=@Id";
            SqlCommand cmd = new SqlCommand(query, con);
            cmd.Parameters.AddWithValue("@Name", name);
            cmd.Parameters.AddWithValue("@Age", age);
            cmd.Parameters.AddWithValue("@Id", id);

            con.Open();
            cmd.ExecuteNonQuery();
            con.Close();

            GridView1.EditIndex = -1;
            BindGrid();
        }
    }

    protected void GridView1_RowDeleting(object sender,
System.Web.UI.WebControls.GridViewDeleteEventArgs e)
    {
        int id = Convert.ToInt32(GridView1.DataKeys[e.RowIndex].Value);

        using (SqlConnection con = new SqlConnection(/* Your connection string here */))
        {
            string query = "DELETE FROM Employees WHERE Id=@Id";
            SqlCommand cmd = new SqlCommand(query, con);
            cmd.Parameters.AddWithValue("@Id", id);

            con.Open();
            cmd.ExecuteNonQuery();
            con.Close();

            BindGrid();
        }
    }

    protected void GridView1_Sorting(object sender,
System.Web.UI.WebControls.GridViewSortEventArgs e)
    {
        string[] sortParts = SortExpression.Split(' ');
```

```
            if (sortParts[0] == e.SortExpression)
            {
                SortExpression = e.SortExpression + (sortParts[1] == "ASC" ? " DESC" : " ASC");
            }
            else
            {
                SortExpression = e.SortExpression + " ASC";
            }
            BindGrid();
        }
    }
}
```

OUTPUT

Add New Employee

Name: mahalakshmi
Age: 18
Department: bca
Add

12.IMPLEMENT THE XML CLASSES

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication20.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title>xml classes example</title>
</head>
<body>
   <form id="form1" runat="server">
   <div style="font_family:Arial">
   <h2>xml classes example</h2>
   <asp:Button ID="btnCreate" runat="server" Text="Create xml file"
OnClick="btnCreate_click" />
   <br /><br />
   <asp:Button ID="btnRead" runat="server" Text="Read xml file" OnClick="btnRead_click" />
   <br /><br />
   <asp:Label ID="lblOutput" runat="server" Text="  " ForeColor="Blue"></asp:Label>
   </div>
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.IO;
using System.Text;
using System.Web.UI;
using System.Xml;

namespace WebApplication20
{
   public partial class WebForm1 : Page
   {
      private string xmlFilePath;

      protected void Page_Load(object sender, EventArgs e)
      {
         xmlFilePath = Server.MapPath("~/App_Data/books.xml");
      }

      protected void btnCreate_click(object sender, EventArgs e)
      {
         try
         {
            XmlDocument doc = new XmlDocument();
            XmlDeclaration xmlDeclaration = doc.CreateXmlDeclaration("1.0", "UTF-8", null);
            doc.AppendChild(xmlDeclaration);
            XmlElement root = doc.CreateElement("books");
            doc.AppendChild(root);
            XmlElement book1 = doc.CreateElement("book");
            book1.SetAttribute("id", "1");

            XmlElement title1 = doc.CreateElement("title");
            title1.InnerText = "The Great Gatsby";

            XmlElement author1 = doc.CreateElement("author");
            author1.InnerText = "F. Scott Fitzgerald";

            book1.AppendChild(title1);
            book1.AppendChild(author1);
            root.AppendChild(book1);
            XmlElement book2 = doc.CreateElement("book");
            book2.SetAttribute("id", "2");

            XmlElement title2 = doc.CreateElement("title");
            title2.InnerText = "1984";

            XmlElement author2 = doc.CreateElement("author");
```

```
            author2.InnerText = "George Orwell";

            book2.AppendChild(title2);
            book2.AppendChild(author2);
            root.AppendChild(book2);
            Directory.CreateDirectory(Server.MapPath("~/App_Data")); // Ensure folder exists
            doc.Save(xmlFilePath);

            lblOutput.Text = "XML file created successfully at: " + xmlFilePath;
        }
        catch (Exception ex)
        {
            lblOutput.Text = "Error creating XML file: " + ex.Message;
        }
    }

    protected void btnRead_click(object sender, EventArgs e)
    {
        try
        {
            if (!File.Exists(xmlFilePath))
            {
                lblOutput.Text = "XML file does not exist. Please create it first.";
                return;
            }

            XmlDocument doc = new XmlDocument();
            doc.Load(xmlFilePath);

            XmlNodeList books = doc.SelectNodes("/books/book");

            StringBuilder sb = new StringBuilder();
            foreach (XmlNode book in books)
            {
                string id = book.Attributes["id"].Value;
                string title = book["title"].InnerText;
                string author = book["author"].InnerText;

                sb.AppendFormat("Book ID: {0}<br/>Title: {1}<br/>Author: {2}<br/><br/>", id,
title, author);
            }

            lblOutput.Text = sb.ToString();
        }
        catch (Exception ex)
        {
```

```
            lblOutput.Text = "Error reading XML file: " + ex.Message;
        }
    }
  }
}
```

OUTPUT

xml classes example

Create xml file

Read xml file

xml classes example

Create xml file

Read xml file

XML file created successfully at: E:\Sandhiya\WebApplication20\WebApplication20\App_Data\books.xml

xml classes example

Create xml file

Read xml file

Book ID: 1
Title: The Great Gatsby
Author: F. Scott Fitzgerald

Book ID: 2
Title: 1984
Author: George Orwell

13.IMPLEMENT AUTHENTICATION-AUTHORIZATION

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="authentication-
authorization13.aspx.cs"
Inherits="authentication_authorization._13.authentication_authorization13" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
                "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head id="Head1" runat="server">
   <title>Login</title>
</head>
<body>
   <form id="form1" runat="server">
      <h2>Login</h2>

      <asp:Label ID="Label1" Text="Username:" runat="server"
AssociatedControlID="txtUsername" />
      <asp:TextBox ID="txtUsername" runat="server" />
      <br /><br />

      <asp:Label ID="Label2" Text="Password:" runat="server"
AssociatedControlID="txtPassword" />
      <asp:TextBox ID="txtPassword" TextMode="Password" runat="server" />
      <br /><br />

      <asp:Button Text="Login" ID="btnLogin" runat="server" OnClick="btnLogin_Click" />
      <br /><br />

      <asp:Label ID="lblMessage" ForeColor="Red" runat="server" />
   </form>
</body>
</html>
```

BEHIND THE CODE:

```
using System;
using System.Web;
using System.Web.UI;

namespace authentication_authorization._13
{
    public partial class authentication_authorization13 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            // Clear any previous messages on first page load
            if (!IsPostBack)
            {
                lblMessage.Text = "";
            }
        }

        protected void btnLogin_Click(object sender, EventArgs e)
        {
            // Hardcoded credentials for demonstration
            string validUsername = "admin";
            string validPassword = "1234";

            string enteredUsername = txtUsername.Text.Trim();
            string enteredPassword = txtPassword.Text.Trim();

            if (enteredUsername == validUsername && enteredPassword == validPassword)
            {
                // Optionally, you can use Session to store user info
                Session["username"] = enteredUsername;

                // Redirect to a protected page after successful login
                Response.Redirect("Welcome.aspx");
            }
            else
            {
                lblMessage.Text = "Invalid username or password.";
            }
        }
    }
}
```

OUTPUT

Login

Username:

Password:

Login

14.Ticket Reservation Using asp.net control program

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication14.WebForm1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head id="Head1" runat="server">
   <title>Ticket Reservation</title>
</head>
<body>
   <form id="form1" runat="server">

      <div style="width: 400px; margin: 30px auto; font-family: Arial, sans-serif;">
         <h2>Ticket Reservation</h2>

         <asp:Label ID="lblName" runat="server" Text="Name:"
AssociatedControlID="txtName" />
         <asp:TextBox ID="txtName" runat="server" Width="250px"></asp:TextBox>
         <br /><br />

         <asp:Label ID="lblEmail" runat="server" Text="Email:"
AssociatedControlID="txtEmail" />
         <asp:TextBox ID="txtEmail" runat="server" Width="250px"></asp:TextBox>
         <br /><br />

         <asp:Label ID="lblTickets" runat="server" Text="Number of Tickets:"
AssociatedControlID="ddlTickets" />
         <asp:DropDownList ID="ddlTickets" runat="server" Width="100px">
            <asp:ListItem>1</asp:ListItem>
            <asp:ListItem>2</asp:ListItem>
            <asp:ListItem>3</asp:ListItem>
            <asp:ListItem>4</asp:ListItem>
            <asp:ListItem>5</asp:ListItem>
         </asp:DropDownList>
         <br /><br />

         <asp:Button ID="btnReserve" runat="server" Text="Reserve Ticket(s)"
OnClick="btnReserve_Click" />
         <br /><br />

         <asp:GridView ID="gvReservations" runat="server" AutoGenerateColumns="false"
            Width="100%" EmptyDataText="No reservations yet."
            onselectedindexchanged="gvReservations_SelectedIndexChanged">
            <Columns>
```

```
                <asp:BoundField HeaderText="Name" DataField="Name" />
                <asp:BoundField HeaderText="Email" DataField="Email" />
                <asp:BoundField HeaderText="Tickets Reserved" DataField="Tickets" />
            </Columns>
        </asp:GridView>
    </div>

  </form>
</body>
</html>
```

C#:

```
using System;
using System.Collections.Generic;
using System.Web.UI;

namespace WebApplication14
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        public class Reservation
        {
            public string Name { get; set; }
            public string Email { get; set; }
            public int Tickets { get; set; }
        }
        private List<Reservation> Reservations
        {
            get
            {
                if (Session["Reservations"] == null)
                {
                    Session["Reservations"] = new List<Reservation>();
                }
                return (List<Reservation>)Session["Reservations"];
            }
            set
            {
                Session["Reservations"] = value;
            }
        }
```

```
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                BindGrid();
            }
        }

        protected void btnReserve_Click(object sender, EventArgs e)
        {
            if (string.IsNullOrWhiteSpace(txtName.Text) ||
string.IsNullOrWhiteSpace(txtEmail.Text))
            {
                return;
            }
            var newReservation = new Reservation
            {
                Name = txtName.Text.Trim(),
                Email = txtEmail.Text.Trim(),
                Tickets = int.Parse(ddlTickets.SelectedValue)
            };
            var list = Reservations;
            list.Add(newReservation);
            Reservations = list;

            // Clear inputs
            txtName.Text = "";
            txtEmail.Text = "";
            ddlTickets.SelectedIndex = 0;
            BindGrid();
        }

        private void BindGrid()
        {
            gvReservations.DataSource = Reservations;
            gvReservations.DataBind();
        }
        protected void gvReservations_SelectedIndexChanged(object sender, EventArgs e)
        {
        }
    }
}
```

OUTPUT:

Ticket Reservation

Name: gayathri

Email: gayathri@gmail.com

Number of Tickets: 1

Reserve Ticket(s)

Name	Email	Tickets Reserved
mahalakshmi	maha@gmail.com	1
Fahmitha	fahmi@gmail.com	1

15.ONLINE EXAMINATION USING ASP.NET CONTROL

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="maha_online_examination_pg15.WebForm1" %>


<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head id="Head1" runat="server">
   <title>Simple Online Exam</title>
</head>
<body>
   <form id="form1" runat="server">
      <asp:PlaceHolder ID="phQuestions" runat="server"></asp:PlaceHolder>
      <br />
      <asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click"
/>
      <br /><br />
      <asp:Label ID="lblResult" runat="server" Font-Bold="true"
ForeColor="Green"></asp:Label>
   </form>
</body>
</html >
```

BEHIND THE CODE:

```
using System;
using System.Collections.Generic;
using System.Web.UI.WebControls;

namespace maha_online_examination_pg15
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        // Question class
        public class Question
        {
            public string Text { get; set; }
            public List<string> Options { get; set; }
            public int CorrectOptionIndex { get; set; }
        }

        // Sample questions
        private List<Question> questions = new List<Question>
        {
            new Question
            {
                Text = "What is the capital of India?",
                Options = new List<string> { "Mumbai", "Delhi", "Chennai", "Kolkata" },
                CorrectOptionIndex = 1
            },
            new Question
            {
                Text = "What is 5 + 7?",
                Options = new List<string> { "10", "11", "12", "13" },
                CorrectOptionIndex = 2
            },
            new Question
            {
                Text = "Which planet is called the Red Planet?",
                Options = new List<string> { "Earth", "Mars", "Jupiter", "Venus" },
                CorrectOptionIndex = 1
            }
        };

        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                DisplayQuestions();
```

```
    }
}

private void DisplayQuestions()
{
    phQuestions.Controls.Clear();

    for (int i = 0; i < questions.Count; i++)
    {
        // Question text label
        Label lblQ = new Label();
        lblQ.Text = "{i + 1}. {questions[i].Text}";
        phQuestions.Controls.Add(lblQ);

        // Add line break
        phQuestions.Controls.Add(new Label { Text = "<br />" });

        // RadioButtonList for options
        RadioButtonList rblOptions = new RadioButtonList();
        rblOptions.ID = "rblQ" + i;

        foreach (var option in questions[i].Options)
        {
            rblOptions.Items.Add(new ListItem(option));
        }

        phQuestions.Controls.Add(rblOptions);

        // Add two line breaks for spacing
        phQuestions.Controls.Add(new Label { Text = "<br />" });
        phQuestions.Controls.Add(new Label { Text = "<br />" });
    }
}

protected void btnSubmit_Click(object sender, EventArgs e)
{
    int score = 0;

    for (int i = 0; i < questions.Count; i++)
    {
        RadioButtonList rbl = (RadioButtonList)phQuestions.FindControl("rblQ" + i);
        if (rbl != null && rbl.SelectedIndex == questions[i].CorrectOptionIndex)
        {
            score++;
        }
    }
```

```
            lblResult.Text = "You scored {score} out of {questions.Count}";
        }
    }
}
```

OUTPUT:

{i + 1}. {questions[i].Text}

- ○ Mumbai
- ○ Delhi
- ○ Chennai
- ○ Kolkata

{i + 1}. {questions[i].Text}

- ○ 10
- ○ 11
- ○ 12
- ○ 13

{i + 1}. {questions[i].Text}

- ○ Earth
- ○ Mars
- ○ Jupiter
- ○ Venus

Submit

Submit

You scored {score} out of {questions.Count}